SSL Certificate Update Runbook

Nginx lb-server / ai-training.cck.rnu.tn

Status	Value
Result	**Completed successfully** - active HTTPS certificate updated and verified externally.
Domain	`ai-training.cck.rnu.tn`
New certificate validity	**Apr 15 2026** to **Jul 14 2026**
Issuer	Let's Encrypt E7
Remote VM	`data-node01` / `193.95.31.101`
Nginx runtime	Docker/Kubernetes lb-service container: `e2e1e2ed5f0a`

Purpose

This runbook documents the exact successful steps performed: copy from the local machine, validate the new certificate, discover the real Nginx configuration and container mount, replace the active files, fix ownership, reload Nginx, verify HTTPS, and organize all SSL material by date and purpose.

Prepared for operational reuse and future certificate renewals.

1. Current SSL/Nginx Architecture

The production certificate is read by Nginx inside the lb-server container, but the files are stored on the host under /opt/lb/conf.

Item	Path / Value
Public endpoint	`https://ai-training.cck.rnu.tn`
Remote VM	`data-node01` / `193.95.31.101`
Nginx container	`e2e1e2ed5f0a` - lb-service container in kube-system
Host mount	`/opt/lb/conf` is mounted into the container as `/usr/local/NSP/etc/nginx`
Active host certificate	`/opt/lb/conf/vhosts/ai/fullchain1.pem`
Active host private key	`/opt/lb/conf/vhosts/ai/privkey1.pem`
Container certificate path	`/usr/local/NSP/etc/nginx/vhosts/ai/fullchain1.pem`
Container private key path	`/usr/local/NSP/etc/nginx/vhosts/ai/privkey1.pem`
Nginx config file	`/opt/lb/conf/vhosts/proxy_balancer.conf`
Reload script	`/home/paas/reload_nginx.sh`

Active Nginx config logic:

```
listen 172.16.82.18:443 ssl;
server_name 193.95.31.101 www.ai-training.cck.rnu.tn training.cck.rnu.tn;

ssl_certificate     vhosts/ai/fullchain1.pem;
ssl_certificate_key vhosts/ai/privkey1.pem;
```

Important rule

Do not move the active files under /opt/lb/conf/vhosts/ai/. The /home/paas/SSL-AI folder is for organization, documentation, backups, and inventory. Production Nginx reads from /opt/lb/conf/vhosts/ai/.

2. Local Certificate Package and Upload

The new certificate files started on the local machine under the Downloads folder.

Local source directory:

```
cd ~/Downloads/cert-ai-training
ls -lh

# Files present locally:
cert.pem
chain.pem
fullchain.pem
privkey.pem
README
```

Verify the certificate before uploading:

```
openssl x509 -in fullchain.pem -noout -subject -issuer -dates

# Expected result:
# subject=CN = ai-training.cck.rnu.tn
# issuer=C = US, O = Let's Encrypt, CN = E7
# notBefore=Apr 15 14:59:43 2026 GMT
# notAfter=Jul 14 14:59:42 2026 GMT
```

Verify that certificate and private key match:

```
echo "CERT HASH:"
openssl x509 -in fullchain.pem -noout -pubkey \
| openssl pkey -pubin -outform der \
| openssl dgst -sha256

echo "KEY HASH:"
openssl pkey -in privkey.pem -pubout -outform der \
| openssl dgst -sha256

# Both hashes matched:
# 8c819859e78a4239236ff1a53bb7db5df9b0ba3e2f6d202bf2393e6ecd65abf3
```

Upload to the remote VM user home:

```
scp fullchain.pem privkey.pem cert.pem chain.pem README \
paas@193.95.31.101:~/
```

Note about the first failed upload

The initial path /paas/SSL-AI/ failed because the real home path is /home/paas. Uploading to paas@193.95.31.101:~/ correctly placed the files in /home/paas/.

3. Discovery of the Real Active Nginx Paths

The host nginx -T did not show SSL config because the real Nginx runs inside the lb-server container.

Find the lb-server / Nginx container:

```
docker ps | grep -i "lb-server\|nginx"

# Container used:
# e2e1e2ed5f0a
```

Inspect container mounts:

```
CID="e2e1e2ed5f0a"

docker inspect "$CID" --format '{{range .Mounts}}{{println .Source " -> " .Destination}}{{end}}'

# Important mount discovered:
# /opt/lb/conf  ->  /usr/local/NSP/etc/nginx
```

Read the active Nginx SSL references from inside the container:

```
docker exec "$CID" bash -lc '
/usr/local/NSP/sbin/nginx -T 2>&1 \
| grep -nE "server_name|listen .*443|ssl_certificate|ssl_certificate_key"
'

# Key output:
# listen 172.16.82.18:443 ssl;
# server_name 193.95.31.101 www.ai-training.cck.rnu.tn training.cck.rnu.tn;
# ssl_certificate vhosts/ai/fullchain1.pem;
# ssl_certificate_key vhosts/ai/privkey1.pem;
```

Conclusion of discovery

The real active host files are /opt/lb/conf/vhosts/ai/fullchain1.pem and /opt/lb/conf/vhosts/ai/privkey1.pem. They appear inside the container as /usr/local/NSP/etc/nginx/vhosts/ai/fullchain1.pem and /usr/local/NSP/etc/nginx/vhosts/ai/privkey1.pem.

4. Backup, Replace, Fix Permissions, Reload

These are the exact working commands used on the remote VM.

Create a backup of the active Nginx certificate pair:

```
cd /home/paas

STAMP="$(date +%Y%m%d_%H%M%S)"
mkdir -p /home/paas/SSL-AI/backup-nginx-active-$STAMP

cp -a /opt/lb/conf/vhosts/ai/fullchain1.pem \
/home/paas/SSL-AI/backup-nginx-active-$STAMP/

cp -a /opt/lb/conf/vhosts/ai/privkey1.pem \
/home/paas/SSL-AI/backup-nginx-active-$STAMP/
```

Replace the active certificate files:

```
install -m 600 /home/paas/fullchain.pem \
/opt/lb/conf/vhosts/ai/fullchain1.pem

install -m 600 /home/paas/privkey.pem \
/opt/lb/conf/vhosts/ai/privkey1.pem
```

Fix ownership for the container user:

```
CID="e2e1e2ed5f0a"

CUID="$(docker exec "$CID" id -u)"
CGID="$(docker exec "$CID" id -g)"

# Discovered:
# uid=10010(lb), gid=10010(lb)

chown "$CUID:$CGID" \
/opt/lb/conf/vhosts/ai/fullchain1.pem \
/opt/lb/conf/vhosts/ai/privkey1.pem

chmod 644 /opt/lb/conf/vhosts/ai/fullchain1.pem
chmod 600 /opt/lb/conf/vhosts/ai/privkey1.pem
```

Test and reload Nginx:

```
docker exec "$CID" bash -lc '/usr/local/NSP/sbin/nginx -t'

cd /home/paas
./reload_nginx.sh

# Successful result:
# nginx.conf syntax is ok
# nginx.conf test is successful
# [INFO] reload_nginx Successful.
```

5. Final Validation

The certificate was verified at file level, container level, Nginx config level, and public HTTPS level.

Validation	Command
Active host certificate date	`openssl x509 -in /opt/lb/conf/vhosts/ai/fullchain1.pem -noout -subject -issuer -dates`
Nginx config test	`docker exec "$CID" bash -lc '/usr/local/NSP/sbin/nginx -t'`
Public HTTPS check	`echo | openssl s_client -connect ai-training.cck.rnu.tn:443 -servername ai-training.cck.rnu.tn 2>/dev/null | openssl x509 -noout -subject -issuer -dates`

Final confirmed result:

```
subject=CN = ai-training.cck.rnu.tn
issuer=C = US, O = Let's Encrypt, CN = E7
notBefore=Apr 15 14:59:43 2026 GMT
notAfter=Jul 14 14:59:42 2026 GMT
```

Operational status

The public endpoint is now serving the new Let's Encrypt E7 certificate. The certificate expires on Jul 14 2026, so renewal should be planned before that date.

6. Final Folder Organization

The /home/paas directory was cleaned. SSL material is now grouped under /home/paas/SSL-AI by purpose and date.

Clean /home/paas top-level view:

```
/home/paas/
|-- check_interface_status.sh
|-- client_software_install/
|-- common/
|-- database_bak/
|-- fix_both.sh
|-- lb.log
|-- reload_nginx.sh
|-- SSL-AI/
`-- system_reinforcement.sh
```

Organized SSL archive architecture:

```
/home/paas/SSL-AI/
|-- 00-ACTIVE-PRODUCTION-DO-NOT-MOVE/
|   |-- active-paths.README.txt
|   `-- 2026-04-15_to_2026-07-14_current-active/
|-- 01-NEW-CERTIFICATES/
|   `-- 2026-04-15_to_2026-07-14_letsencrypt_E7/
|       |-- cert.pem
|       |-- chain.pem
|       |-- fullchain.pem
|       |-- privkey.pem
|       `-- README
|-- 02-EXPIRED-CERTIFICATES/
|   `-- 2025-10-21_legacy-server-certs/
|-- 03-OLD-NGINX-BACKUPS/
|   `-- 2025-07_nginx-backup-files/
|-- 04-CONFIG-BACKUPS/
|   `-- proxy-balancer/
|-- 05-INVENTORY/
|   |-- cert_inventory.sh
|   |-- cert_inventory_20260427_190503.txt
|   `-- opt-lb-conf-vhosts-ai-copy/
|-- backup-20260427_183256/
|-- backup-active-20260427_183436/
`-- backup-nginx-active-20260427_184226/
```

Organization principle

Folder names use the certificate validity period when known, for example 2026-04-15_to_2026-07-14. This is more reliable than trusting generic filenames like server.pem or certificate.pem.

7. Inventory, Maintenance, and Rollback

Use these commands for future renewal checks, documentation, and emergency rollback.

Generate certificate inventory with dates:

```
/home/paas/SSL-AI/05-INVENTORY/cert_inventory.sh \
> /home/paas/SSL-AI/05-INVENTORY/cert_inventory_$(date +%Y%m%d_%H%M%S).txt

tail -n +1 /home/paas/SSL-AI/05-INVENTORY/cert_inventory_*.txt
```

Future renewal verification checklist:

```
openssl x509 -in /opt/lb/conf/vhosts/ai/fullchain1.pem \
-noout -subject -issuer -dates

CID="e2e1e2ed5f0a"
docker exec "$CID" bash -lc '/usr/local/NSP/sbin/nginx -t'

echo | openssl s_client \
-connect ai-training.cck.rnu.tn:443 \
-servername ai-training.cck.rnu.tn 2>/dev/null \
| openssl x509 -noout -subject -issuer -dates
```

Emergency rollback pattern:

```
# Choose the desired backup folder, for example:
BACKUP="/home/paas/SSL-AI/backup-nginx-active-20260427_184226"

install -m 600 "$BACKUP/fullchain1.pem" \
/opt/lb/conf/vhosts/ai/fullchain1.pem

install -m 600 "$BACKUP/privkey1.pem" \
/opt/lb/conf/vhosts/ai/privkey1.pem

chown 10010:10010 \
/opt/lb/conf/vhosts/ai/fullchain1.pem \
/opt/lb/conf/vhosts/ai/privkey1.pem

chmod 644 /opt/lb/conf/vhosts/ai/fullchain1.pem
chmod 600 /opt/lb/conf/vhosts/ai/privkey1.pem

docker exec "$CID" bash -lc '/usr/local/NSP/sbin/nginx -t'
cd /home/paas && ./reload_nginx.sh
```

Final reminder

When renewing next time, update only the active pair under /opt/lb/conf/vhosts/ai/, keep ownership as 10010:10010, keep fullchain1.pem readable and privkey1.pem private, then test and reload.